
HOSPITAL STAFF REFERENCE

HospitalOS

Tenant Guide

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AgoroX Africa · HospitalOS

Hospital Operating System for Africa

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1. Who this guide is for

This guide is for hospital staff using HospitalOS day to day — clinicians, diagnostic and pharmacy staff, finance and administration teams. It covers every module in the system and, where it matters, why a screen behaves the way it does.

Role	Scope	Account
Administrator	Your hospital's full HospitalOS deployment	Any account with the Super Admin role
Clinical staff	Patient care, diagnostics, pharmacy for your role	Doctor / Nurse / Lab Scientist / Radiographer / Pharmacist
Finance staff	Billing, payments, claims	Cashier account

Two-level permissions.

Every role has an AREA list (which sidebar groups it can open) and a GRANTS list (which specific actions it can take once inside). Section 3 covers this in full, with the actual permission table.

2. Signing in and how the system is organised

Sign in with your hospital email. An administrator sets your role under Administration → Users & roles. Authentication is verified server-side against securely hashed credentials — never checked in the browser.

2.1 The fifteen sidebar groups

The sidebar is arranged by workflow, roughly in the order work happens through a hospital, not by department hierarchy:

Group	What lives there
Overview	Dashboard, alerts, worklist, communication hub, global search
Patient care	Registration/ADT, wards, ICU, emergency, maternity, records, renal, bookings, referrals
Laboratory	Laboratory (291 tests, 12 disciplines), point of care testing, lab utilities, biobanking
Diagnostics	Blood bank, imaging, radiotherapy, diagnostic intelligence, instruments & Gateway Agent
Pharmacy	Dispensing, inventory, formulary
Specialty services	21 specialist clinics (11 medical, 10 surgical) as one filterable module, plus dental & oral health

Group	What lives there
Finance & trade	Billing, payments, insurance & NHIA claims
Accounts & Procurement	Procurement, stores & assets, bank reconciliation
Operations	Scheduling, CSSD, biomedical equipment, fleet, facility, sites, support, visitors
Academic	Training, logbooks, CME, research, ethics committee
Public health	Surveillance, immunisation, outreach, reporting
Intelligence	Analytics, reports, forecasting
Compliance	Practitioner licenses, facility accreditation, incident & risk, policies & SOPs, account security
Administration	Users & roles, facilities & sites, pricing, lab reference ranges, clinical decision rules, documents, audit, settings
Academy	AgoroX certification and training content

Two groups split out, deliberately.

Laboratory used to sit inside Diagnostics — fine at 41 tests, unwieldy once the catalogue grew to 291 across 12 disciplines. Accounts & Procurement split out of Finance for a sharper reason: a cashier holding cash-handling access alongside procurement and bank reconciliation access is a real internal-controls weakness, not just a navigation inconvenience — the same separation-of-duties principle behind a nurse not being able to discharge a patient.

You only see what your role grants.

A pharmacist's sidebar shows Overview and Pharmacy; a cashier's shows Overview and Finance, but not Accounts & Procurement — that now belongs to a separate Accountant role. This is not secrecy — it keeps each person's view to what they actually use.

3. Roles and permissions

Area permissions gate the sidebar and routes. Action permissions gate the buttons within an area. Separation of duties is enforced, not just suggested:

Role	Can	Cannot
Nurse	Admit patients, record vitals, file clinical notes	Discharge a patient
Lab Scientist	Collect samples, enter results, verify results	Reach Patient care at all
Radiographer	Order and report imaging studies	Verify laboratory results
Pharmacist	Dispense medication, restock inventory	Reach Diagnostics at all
Cashier	Take payments, file insurance claims	Approve a claim, reach Accounts & Procurement
Accountant	Procurement, stores, bank reconciliation	Take payments or handle cash
Records Officer	Register new patients	Admit or discharge a patient
Operations Manager	Biomedical, fleet, CSSD, facilities	Reach clinical or financial areas
Compliance Officer	Accreditation, incidents, policies	Reach clinical or financial areas
Public Health Officer	Surveillance, immunisation, reporting	Reach clinical or financial areas
Doctor	Full patient-care and diagnostics actions, override a clinical-rule block	Administration functions
Super Admin	Everything	—

Cashier and Accountant are deliberately separate.

A cashier who could also approve purchase orders and reconcile the bank statement would be able to cover their own tracks — the same reasoning behind a cashier being unable to approve their own claims.

The last-admin protection.

The last active Super Admin account cannot be deactivated by anyone, including another Super Admin — this prevents a hospital from accidentally locking itself out of its own system.

4. Overview and the alert system

The Dashboard opens on live occupancy, admissions, and revenue charts. Every stat card links directly into the module behind it. Press Ctrl+K (Cmd+K on Mac) from anywhere to search patients, staff, results, medicines, invoices, payments, appointments, procedures, screens, and documentation at once — see Section 11.

4.1 Alerts — twenty-three sources, one feed

Source	Raises an alert when...	Severity
Laboratory	A result crosses a panic threshold	Critical
Critical care	Any ICU/HDU vital goes critical	Critical
Radiology	A report is flagged with an urgent finding	Critical
Maternity	A newborn's Apgar score is below 7	Critical
Blood bank	A group is below reorder / a unit nears expiry	Critical / Warning
Pharmacy	A drug is out of stock / below reorder	Critical / Warning
Instruments	A device errors or goes offline	Critical / Warning
Operations	Equipment under repair / a vehicle out of service	Warning
Oncology	A chemotherapy cycle is overdue	Warning
Public health	A notifiable disease is trending upward	Warning
Renal	A scheduled dialysis session is overdue	Critical
Referrals	Emergency-urgency inbound referral logged	Critical
Immunisation	A scheduled dose is 14+ days overdue	Warning
Privacy	A data-subject request is past the 30-day window	Critical
Sickle Cell Centre	An active severe crisis	Critical
Infection Prevention & Control	3+ open cases of the same infection type	Critical
Geriatric Unit	High falls risk or polypharmacy (5+ medications)	Warning
Mental Health Unit	Constant observation or an active risk flag	Critical
Compliance	A practitioner licence expired or expires within 60 days	Critical / Warning
Compliance	A facility accreditation expired or expires within 60 days	Critical / Warning
Incident & Risk	A Severe harm or Sentinel event is still open	Critical

Source	Raises an alert when...	Severity
Policies & SOPs	A policy's review date has passed or is within 60 days	Warning
Patient journeys	A required step sits unfulfilled over 2 hours	Warning

Most alerts clear themselves — they read live state, so fixing the underlying problem clears the alert automatically, nothing to dismiss.

4.2 Notifications

Not yet functional as email/SMS push.

The bell icon currently surfaces booking requests inside the app only. It does not yet push a notification outside HospitalOS — that requires the Communication hub's channels to be connected to a real SMS/email gateway. Its count does not shrink on its own; a booking only leaves the bell once someone actually confirms or declines it.

5. Patient care

5.1 Registration, admission and the ward board

The bed board and ADT read the same registry, so a bed taken in one is unavailable in the other instantly — double-booking is not possible.

5.2 Accommodation tiers

Tier	Rate / night
General Ward	₦15,000
Semi-Private	₦35,000
Private Room	₦60,000
Private Suite	₦120,000
VIP Suite	₦220,000
Executive Suite	₦350,000
Critical Care	₦180,000

Catalogue defaults — every rate can be overridden for your hospital. See Section 9.1.

5.3 The medical record

SOAP-structured clinical notes, an ICD-10 problem list, and recorded allergies. Filed notes cannot be edited or deleted — a correction is filed as an amendment referencing the original, with both remaining visible.

Allergy safety is enforced, not advisory.

A severe allergy on a patient's record disables the Pharmacy dispense button outright for that drug — there is no override in the interface.

Clinical decision rules add a second, configurable layer.

Beyond allergy blocking, your hospital can now configure its own real-time checks — an advisory, a warning requiring confirmation, or a hard block — on prescribing, scheduling a procedure, or ordering a lab test. See Section 9.3.

5.4 Referrals

Referrals in and out of your facility are tracked from first contact through to acceptance or decline, and through to the patient's arrival.

5.5 Renal & dialysis, ICU/HDU, Emergency, Maternity, Online bookings

- **Renal & dialysis** runs a haemodialysis programme (vascular access, schedule, per-session fluid removal) and a CKD registry that auto-stages from eGFR.
- **ICU/HDU** tracks five vitals per bed with panic thresholds; a critical reading marks the patient Unstable and raises an alert automatically.
- **Emergency** sorts by triage acuity first, arrival time second, and now silently tracks each patient's journey through discharge — see Section 9.4.
- **Maternity** tracks labour through to delivery; a newborn Apgar below 7 raises a critical alert immediately.
- **Online bookings** holds appointment requests from your website. Confirming and checking in a booking calls the same function Outpatient uses internally.

5.6 Specialist clinics and rehabilitation

Specialty services → Specialist clinics covers the full range a tertiary hospital runs — 11 medical specialties (including Geriatrics) and 10 surgical specialties, 21 in total — as one module with a department filter, rather than a separate sidebar entry per specialty. A cardiologist and a nephrologist both run a clinic, see referrals, and manage a patient list; the department is context you filter by, not a different piece of software.

Rehabilitation & therapy runs alongside it for physiotherapy, occupational therapy, and speech therapy referrals, tracked as a course of sessions against a treatment plan.

Dental & oral health checks patients through Waiting → In chair → Completed. Unlike a medical specialty clinic, there is no diagnosis or problem-list tracking — the visit record is the set of procedures performed (oral examination, scaling & polishing, filling, extraction, root canal treatment, dental X-ray, crown fitting, denture fitting), matching how dental billing genuinely works.

6. Laboratory & Diagnostics

Laboratory and Diagnostics are two separate top-level sidebar groups as of this update — documented together here since the content is closely related, not because one sits inside the other.

291 laboratory tests across twelve disciplines, and 27 imaging protocols across Ultrasound, CT, MRI, General Radiography, and Mammography.

6.1 Laboratory

Orders move Ordered → Collected → Resulted → Verified; each result flags Low/High/Critical live against its reference range — your hospital's own configured range if you've set one, the catalogue default otherwise. See Section 9.2.

6.2 Instruments & devices gateway

One interoperability hub, reachable from both Diagnostics and Administration:

Category	Protocol	Examples
Laboratory analyzer	HL7 v2/MLLP or ASTM E1381/E1394, auto-detected	Sysmex, Cobas, a 2009 analyzer still in service
Imaging modality	DICOM 3.0	CT, MRI, ultrasound, a 2007 CR reader
Radiotherapy system	DICOM-RT / proprietary	A modern LINAC and a manual Cobalt-60 unit
Printer	IPP / raw socket / serial	Label, report, and legacy dot-matrix printers

Real, live connections — via the Gateway Agent.

Register a device here, and a device token appears once — copy it immediately, it can't be retrieved again. Contact AgoroX with the device name to arrange installing the Gateway Agent, a small program on a PC next to the equipment, currently a hands-on setup step rather than a self-service download. The agent auto-detects HL7/MLLP vs. ASTM E1381/E1394 — no manual protocol configuration — and queues a result locally if your internet drops, sending it once connectivity returns.

DICOM and proprietary protocols aren't supported by the agent yet.

Imaging (DICOM) is a substantially larger undertaking than HL7/ASTM, and a fully proprietary protocol without a public spec can't be connected without that vendor's cooperation. Each 'receive/confirm/send' action on this screen still runs the exact code path a real listener would call — useful for testing before the physical agent is installed.

6.3 Biobanking & lab utilities

Biobanking is a separate concern from the Laboratory worklist — specimens retained after routine testing for research, future clinical use, or medico-legal purposes, not samples awaiting a result today.

Storage spans four capacity-limited units — two freezers, a liquid-nitrogen vault, and a room-temperature archive — each tracking specimen type, volume, consent basis (research use, future clinical use only, or no further use), and associated study. Lab utilities separately holds clinical calculators, unit converters, and bench reference material.

A full unit refuses new specimens.

Banking a specimen checks the destination unit's capacity before accepting it — rather than silently overfilling.

6.4 Blood bank

Diagnostics → Blood bank tracks inventory by unit and processes crossmatch requests against real ABO/Rh compatibility logic — not a simplified same-group-only rule. Each recipient blood group maps to a specific list of compatible donor groups following standard transfusion medicine rules (O-negative as the universal donor, AB-positive as the universal recipient). A crossmatch automatically reserves a compatible available unit; if none exists, the request is refused outright rather than silently reserving an incompatible one.

Batch delivery, individual tracking.

Adding stock lets you set a blood group, expiry window, and quantity (up to 100) at once — each unit still gets its own sequential tag and appears individually in inventory. A transfused unit becomes one of the five automatic billing sources, charged the moment the transfusion is recorded.

The two failure modes that actually matter.

A group below its reorder threshold, or a unit approaching expiry, both raise alerts — running out and wasting expired stock.

7. Pharmacy

Dispensing checks recorded allergies before completing a transaction and blocks any quantity greater than available stock. Inventory alerts clear themselves once stock is restocked above the reorder level.

8. Finance & trade

Billing aggregates charges from five sources automatically: laboratory, pharmacy, radiology, theatre, and accommodation. A theatre case becomes billable the moment it enters theatre, not

when scheduled. Claims move Submitted → Approved/Rejected → Paid — a cashier can file a claim but cannot approve it. Payment collection is processed through your hospital's configured gateway.

9. Making HospitalOS your own

9.1 Setting your own prices

Administration → Pricing lets you override the catalogue default for any test, drug, imaging study, procedure, or accommodation tier. Every billing calculation and every price shown to staff reads through the same value.

Past charges are never rewritten.

A patient billed before you changed a price keeps their original total; only new transactions use the new price.

9.2 Setting your own lab reference ranges

Administration → Lab reference ranges works the same way as Pricing: set your own normal and critical ranges per test and analyte, or leave the catalogue default in place. Every result flags against your range immediately, and only new results use a changed range — past results keep whatever flag they were given at the time.

9.3 Configuring clinical decision rules

Administration → Clinical decision rules is where your hospital writes and reviews its own real-time safety checks — which action triggers a rule (prescribing, scheduling a procedure, ordering a lab test), what condition it checks, and what severity it carries: a quiet advisory, a warning requiring confirmation, or a hard block only a doctor or administrator can override. Every evaluation is logged and reviewable.

9.4 Patient journeys

The Emergency department automatically starts a silent journey for every patient presented — no extra step required. It tracks Triage → Lab (optional) → Pharmacy (optional) → Discharge, and flags under Alerts if a required step sits unfulfilled too long. A medication only fulfils its step when actually dispensed, not merely prescribed — catching exactly the gap where a patient is triaged but never actually receives what was ordered.

9.5 Adding to your equipment, fleet, and site registers

Operations → Biomedical, Operations → Ambulance & fleet, and Administration → Facilities & sites can each be added to directly. New equipment gets an auto-generated tag; a new vehicle's registration number is entered by you, since it's a real external identifier you already have. Every addition writes a full audit trail entry.

9.6 Your branding

Administration → Settings has a hospital name and logo URL field. Both appear top-right on every screen, live, within seconds of saving — and carry onto every printed lab result, imaging report, and receipt.

9.7 Your documents

Administration → Documents & templates accepts real uploads, stored in Cloudflare R2 — drag a file in, assign any category you like, and it is immediately downloadable, renameable, and deletable, and persists properly across sessions.

10. The audit trail

Administration → Security & audit is append-only by construction: entries are frozen the moment they are written, with no update or delete API at all, stored server-side in the production database. Each entry is hash-chained to the one before it, so editing a past record breaks verification at that record, and deleting one breaks verification at the next.

This is a genuine, enforceable control today.

Tamper-evident means altering or deleting a past entry breaks the chain visibly — not a theoretical property waiting on a future migration, since the chain already lives in the real production database, not the browser.

11. Compliance & risk

Three sub-modules under the Compliance nav group cover the legal basis for operating and how your hospital manages risk — tracked with real expiry dates and real audit trails, not filed away and forgotten.

11.1 Practitioner licenses and facility accreditation

Every clinical practitioner's professional license, your facility's own accreditation, and a log of regulatory inspections. Adding a license picks the actual staff member from your hospital's own account list, not a free-typed name — a license can never end up attached to someone without an actual account.

Role	Licensing body
Doctor	MDCN — Medical and Dental Council of Nigeria
Nurse	NMCN — Nursing and Midwifery Council of Nigeria
Lab Scientist	MLSCN — Medical Laboratory Science Council of Nigeria
Radiographer	RRBN — Radiographers Registration Board of Nigeria
Pharmacist	PCN — Pharmacists Council of Nigeria

Facility accreditation covers State Ministry of Health licenses, HEFAMAA registration, NHIA facility accreditation, and laboratory-specific accreditation like ISO 15189 — all tracked the same way: a certificate number, an issue date, and an expiry date.

60 days out, then expired — the same alert system, not a separate reminder.

Both licenses and accreditations raise a hospital-wide alert once within 60 days of expiry, and a more severe one once actually expired.

11.2 Incident and risk management

A structured log for adverse events and sentinel events — what happened, its severity, and its investigation status. A Severe harm or Sentinel event incident left open raises a critical, hospital-wide alert until it's genuinely closed out, not just filed.

11.3 Policies and SOPs

Policy documents with their own review cycle — a policy whose scheduled review date has passed, or is within 60 days, raises a warning, the same pattern as license and accreditation expiry.

12. Getting help

Press Ctrl+K (Cmd+K on Mac) from anywhere to search patients, doctors, laboratory results, medicines, invoices, payments, appointments, procedures, screens, and the full help library — nine result types, one query, each respecting what your role can actually see. Help & documentation is pinned at the bottom of the sidebar, outside the workflow groups, reflecting that it is a standalone reference.

12.1 Keyboard shortcuts

Action	How
Search	Ctrl+K (Cmd+K on Mac) — opens the command palette from anywhere
Close a modal or the search palette	Escape
Move between search results	Arrow keys, then Enter to open the highlighted one
Expand / collapse a nav group	Click the group header in the sidebar
Jump to a module from the Dashboard	Click any stat card
Open the mobile menu	Tap the hamburger icon in the topbar (under 900px width)

Note: the Overview group in the sidebar cannot be collapsed — Dashboard, Alerts, and the Communication hub are meant to be one click away at all times.

12.2 Downloading this guide

This guide is available at any time from Help & documentation — click “Download the full Tenant Guide” on the landing page.

12.3 Frequently asked questions

Why does the sidebar look different for me than for my colleague?

You each see only the groups your role grants — a pharmacist's sidebar and a cashier's sidebar are genuinely different, not a display preference.

Why doesn't the patient get a text message when their result is ready?

Verifying a result and releasing it are two separate steps. Releasing queues a message to the patient through Communication — but actual SMS/WhatsApp delivery isn't wired to a real carrier yet, so the message queues genuinely but won't yet reach a phone.

Is the audit log tamper-proof?

It is tamper-evident, the honest and standard claim for this kind of log: the hash chain lives in the database, not the browser, so tampering after the fact becomes detectable — not that tampering is physically impossible for anyone with database access.

Does HospitalOS support NHIA or private HMOs?

Both. Insurance & NHIA claims tracks NHIA (formerly NHIS) alongside several private HMOs, through the same Submitted → Approved/Rejected → Paid lifecycle.

Why can't I take a cash payment?

Cash specifically requires an open cash session, since it's the only method that touches a physical drawer — open one from Finance → Payments with a counted opening balance first. Every other payment method works without one.

Can two people use the same login at once?

Sessions are tracked server-side. Concurrent-session limiting per account isn't enforced yet, so two people signing in with the same credentials both currently get a valid session — sharing a login is a genuine accountability risk regardless, since every action is attributed to that one account in the audit trail.

13. Known limitations checklist

An honest account, deliberately short — everything not listed here is real.

No real SMS/WhatsApp/email delivery yet.

The Communication hub's queue, templates, and channel selection are all real and persist. What isn't wired up is an actual carrier — messages are genuinely logged but stay at “queued” rather than reporting real delivery status.

The Gateway Agent covers HL7/ASTM, not DICOM or proprietary protocols.

Real, live analyzer connections exist — not simulated — for HL7 v2/MLLP and ASTM E1381/E1394, via a small program installed on a PC at your hospital. DICOM (imaging) and fully proprietary vendor protocols aren't supported by the agent yet.

No self-service password reset.

Your hospital's own administrator resets access under Administration → Users & roles — there is no automated “forgot password” email flow yet.

Questions about this guide?

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